No.

Date:

LATIHAN V

Bag A

1. Sederhanakan bilangan imajiner berikut !

   a). j^3 b). j^4 c). j^5

2. Tunjukkan bilangan kompleks berikut dalam diagram argand !

   a). $3 + j6$ b). $2 - j3$ c). $-4 + j6$

3. Tulislah bentuk sekawan (conyugate) dan lawan bil. kompleks berikut

   a). $4 + j$ b). $-4 - j7$ c). $3 - j2$

4. Nyatakan bilangan kompleks berikut dalam bentuk polar

   a). $-1 + j$ b). $-1 + j$ c). $-3 - j2$

5. Nyatakan dalam bentuk cartesian (rectangular

   a). $32\left(\cos\frac{3\pi}{4} + \sin\frac{3\pi}{4}\right)$ b). $4 < \frac{5\pi}{3}$ c). $8 \angle 110^\circ$

Bag B

1. Operasikan perkalian berikut dalam bentuk cartesian

   a). $(6-j)(j5)$ b). $(2-j)(3+j)$ c). $(-2+j3)(-1-2j)$

2. operasikan perkalian berikut dalam bentuk cartesian

   a). $\frac{3+j4}{1-j2}$ b). $\frac{1}{2+j3} + \frac{1}{2-j3}$ c). $\frac{1}{3+j} - \frac{1}{1+j1}$

3. Selesaikan :

   a). $(12+j) - (4+j5)$ b). $(3+j2)(4-j3)$ c). $(2-j5)^2$

4. Diketahui $z_1 = 3 + j2$, $z_2 = 2 - j3$ dan $z_3 = -4 + j$

   Hitung : a). $z_1 z_2$ b). $z_2 z_1 z_3$ c). $\frac{z_3}{z_2}$

Jawaban

Bag A

1. a $j^3 = j^2 j$

   $= -1 \cdot j = -j$

   b. $j^4 = j^2 j^2$

   $= (-1)(-1) = 1$

   c). $j^5 = j^4 \cdot j$

   $= 1 \cdot j$

   $= j$

2. a $3 + j6$

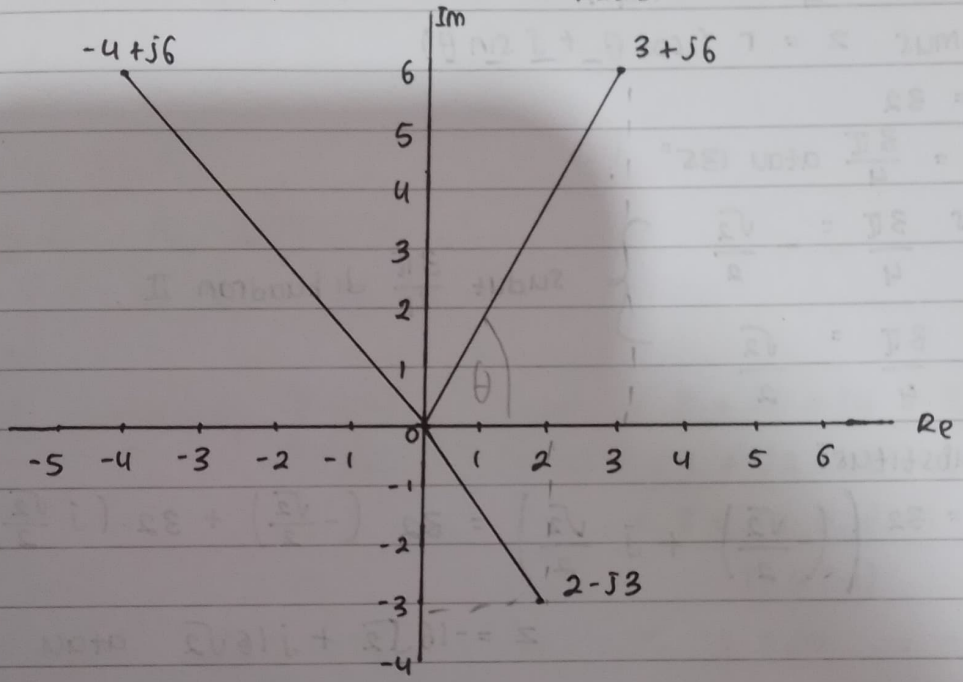
$Re(z) = 3$ dan $Im(z) = 6$ di kuadran I

b $2 - j3$

$Re(z) = 2$ dan $Im(z) = -3$ dan 2 di kuadran IV

c. $-4 + j6$

$Re(z) = -4$ dan $Im(z) = 6$ dan 2 di kuadran II



3. a $4 + j$

sekawan : $4 - j$

lawan : $-4 - j$

b). $-4 - j7$

sekawan : $-4 + j7$

lawan : $4 + j7$

c). $3 - j2$

sekawan : $3 + j2$

lawan : $-3 + j2$

4. a dan b $= -1 + j$

$r = \sqrt{a^2 + b^2}$

$= \sqrt{(-1)^2 + 1^2}$

$= \sqrt{1 + 1} = \sqrt{2}$

$\theta = \arctan \frac{b}{a}$

$\theta = \arctan \left(\frac{1}{-1}\right)$

$= \arctan -1 = \frac{3\pi}{4}$ rad

$= 135°$

$\theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$ rad

dalam bentuk polar adalah $z = \sqrt{2} < \frac{3\pi}{4}$ atau $z = \sqrt{2} < 135$

atau $r(\cos\theta + j\sin\theta) = \sqrt{2}\left(\cos\frac{3\pi}{4} + j\sin\frac{3\pi}{4}\right)$

c. $-3 - j2$

$r = \sqrt{(-3)^2 + (-2)^2}$

$= \sqrt{9 + 4}$

$= \sqrt{13}$

$\theta = \arctan\left(\frac{-2}{-3}\right)$

$= \arctan\left(\frac{2}{3}\right)$

$= 33,69^\circ$

$\theta = \pi + 33,69^\circ$

$= 213,69^\circ$

$= 3,73$ rad

Dalam bentuk polar : $\sqrt{13} < 213,69$ atau $\sqrt{13} < 3,728$

5. a. $32\left(\cos\frac{3\pi}{4} + j\sin\frac{3\pi}{4}\right)$

rumus $z = r(\cos\theta + j\sin\theta)$

$r = 32$

$\theta = \frac{3\pi}{4}$ atau 135°

$\cos\frac{3\pi}{4} = -\frac{\sqrt{2}}{2}$

$\sin\frac{3\pi}{4} = \frac{\sqrt{2}}{2}$

} sudut $\frac{3\pi}{4}$ di kuadran II

substitusi

$z = 32\left(\left(-\frac{\sqrt{2}}{2}\right) + j\frac{\sqrt{2}}{2}\right) = 32\left(-\frac{\sqrt{2}}{2}\right) + 32\left(j\frac{\sqrt{2}}{2}\right)$

$z = -16\sqrt{2} + j16\sqrt{2}$ atau

$-22,627 + j\,22,627$

b. $4 < \frac{5\pi}{3}$

$r = 4$

$\theta = \frac{5\pi}{3}$ atau 300°

$\cos\frac{5\pi}{3} = \frac{1}{2}$

$\sin\frac{5\pi}{3} = -\frac{\sqrt{3}}{2}$

$z = 4\left(\frac{1}{2} + j\left(-\frac{\sqrt{3}}{2}\right)\right) = 4\left(\frac{1}{2}\right) + 4\left(j\left(-\frac{\sqrt{3}}{2}\right)\right)$

$z = 2 - j\,2\sqrt{3}$ atau $2 - j\,3,464$

c. $8 < 110^\circ$

$r = 8$

$\theta = 110^\circ$

$\cos 110^\circ = -0,342$

$\sin 110^\circ = 0,940$

$z = r(\cos\theta + j\sin\theta)$

$z = 8(-0,342 + j.\,0,940)$

$= -2,736 + j\,7,52$

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Bag B

1. a. $(6-j)(j5)$

$6j5 - 5j^2$

$j30 - 5(-1) = 5 + j30$

b $(2-j)(3+j)$

$= 6 + j2 - j3 - j^2$

$= 6 - j - (-1)$

$= 7 - j$

c. $(-2+j3)(-1-j2)$

$= 2 + 2j2 - j3 - 6j^2 \rightarrow -1$

$= 2 + j + 6$

$= 8 + j$

2. a $\frac{3+j4}{1-j2}$ disamakan $\frac{(3+j4)}{(1-j2)} \times \frac{(1+j2)}{(1+j2)} = \frac{3 + j6 + j4 + 8j^2}{1 + j2 - j2 - 4j^2}$

$= \frac{3 + j10 + 8(-1)}{1 - 4(-1)}$

$= \frac{-5 + j10}{5} = -1 + j2$

b. $\frac{1}{2+j3} + \frac{1}{2-j3}$ disamakan penyebutnya

- $\frac{1}{(2+j3)} \times \frac{(2-j3)}{(2-j3)} = \frac{2-j3}{4 - j6 + j6 - 9j^2}$

$= \frac{2-j3}{4-9(-1)}$

$= \frac{2-j3}{4+9} \cdot \frac{2-j3}{13}$

- $\frac{1}{(2+j3)} \times \frac{(2-j3)}{(2-j3)} = \frac{2-j3}{4 - j6 + j6 - 9j2}$

$= \frac{2-j3}{4-9(-1)} = \frac{2-j3}{4+9} = \frac{2-j3}{13}$

$$\frac{(2-j3)}{13} + \frac{(2+j3)}{13} = \frac{4}{13}$$

c. $\frac{1}{3+j} - \frac{1}{1+j7}$

- $\frac{1}{(3+j)} \times \frac{(3-j)}{(3-j)} = \frac{3-j}{9-j3+j3-j^2} = \frac{3-j}{9-(-1)}$

$$= \frac{3-j}{10}$$

- $\frac{1}{(1+j7)} \times \frac{(1-j7)}{(1-j7)} = \frac{1-j7}{1-j7+j7-49j^2} = \frac{1-j7}{1-49(-1)} = \frac{1-j7}{1+49}$

$$= \frac{1-j7}{50}$$

$$\frac{3-j}{10} - \frac{1-j7}{50} = \frac{(15-5j)-(1-j7)}{50} = \frac{14+j2}{50} = \frac{7+j}{25}$$

3. a $(12+j)-(4+j5)$

$= 12+j-4-j5 = 8-j4$

b $(3+j2)(4-j3)$

$= 12-j9+j8-6j^2$

$= 12-j-6(-1)$

$= 12-j+6 = 18-j$

c $(2-j5)^2$

$(2-j5)(2-j5)$

$= 4-j10-j10+25j^2$

$= 4-j20+25(-1)$

$= 4-j20-25$

$= -21-j20$

4. a $z_1 z_2$

$z_1 z_2 = (3+j2)(2-j3)$

$= 6-j9+j4-6j^2$

$= 6-j5-6(-1)$

$= 6-j5+6 = 12-j5$

b. $z_1 z_2 z_3$

$= (2-j3)(3+j2)(-4+j)$

$= (12-j5)(-4+j)$

$= -48 + j12 + j20 - 5j^2$

$= -48 + j32 - 5(-1)$

$= -48 + j32 + 5 \quad = -43 + j32$

c. $\frac{z_3}{z_2}$

$= \frac{(-4+j)}{(3+j2)} \times \frac{(3-j2)}{(3-j2)} = \frac{-12 + j8 + j3 - 2j^2}{9 - \cancel{j6} + \cancel{j6} - 4j^2}$

$= \frac{-12 + j11 - 2(-1)}{9 - 4(-1)}$

$= \frac{-12 + j11 + 2}{9 + 4}$

$= \frac{-10 + j11}{13}$

$= \frac{-10}{13} + j\frac{11}{13}$